Lab: Thought Experiment — Drawing Boundaries

Objective

Explore the philosophical implications of system boundaries by constructing and analyzing thought experiments about where systems begin and end.

Prerequisites

- Familiarity with the concepts of Markov Blankets, autopoiesis, and the life-mind continuity thesis from this module's lecture.

Part 1: The Boundary Problem

Goal: Identify the assumptions hidden in everyday boundary attributions.

Consider three candidate systems: (a) a single cell, (b) a human being walking through a city, (c) a jazz quartet improvising a piece. For each, answer:

- Where would you intuitively place the boundary between the system and its environment?
- What criteria did you use (spatial, functional, informational, phenomenological)?

Write your response here

Part 2: Applying the Markov Blanket

Goal: Translate the intuitive boundary into the formal language of Active Inference.

For the human walking through a city (case b above), identify plausible candidates for:

- **Sensory states (s)**: What aspects of the environment does the person sample?
- **Active states (a)**: What actions does the person take that change the environment?
- **Internal states (μ)**: What hidden states are shielded from the environment by the blanket?
- **External states (η)**: What environmental states are relevant but only known through the blanket?

Does the blanket fall at the skin? Or does it extend further — to include the phone in the person's hand, the headphones, the navigation app?

Write your response here

Part 3: Life-Mind Continuity Challenge

Goal: Construct an argument for or against attributing "mind" to a minimal living system.

Consider a bacterium performing chemotaxis — swimming toward nutrients and away from toxins. The bacterium has internal metabolic states, a membrane (blanket), and active locomotion.

- In what sense, if any, does the bacterium "perceive" its environment?
- Does it have "beliefs" about where nutrients are? Or is this merely a useful metaphor?
- Write a short argument (3-5 sentences) either defending or criticizing the attribution of mind to the bacterium.

Write your response here

Part 4: The Ship of Theseus Revisited

Goal: Apply the Markov Blanket perspective to a classical philosophical puzzle.

A ship has every plank replaced, one by one, over the course of a decade. The old planks are assembled into a second ship. Which is the Ship of Theseus?

Reframe this puzzle using the concept of Markov Blanket dynamics. Is identity a property of material constitution or of dynamical pattern? Does the Markov Blanket framework dissolve the paradox, or does it merely relocate it?

Write your response here

Part 5: Synthesis and Peer Review

Goal: Develop and defend your own philosophical position.

Write a 200-word position statement on one of the following: * "Markov Blankets are real boundaries in nature, not mere mathematical conveniences." * "The life-mind continuity thesis is a category error dressed in formal clothing." * "The Free Energy Principle is best understood as a principle of organization, not a law of physics."

Share your position with a peer. Identify the strongest objection to your view and respond to it.

Write your response here

Lab Summary

Part	Skill Developed	Key Philosophical Move
1	Boundary identification	Exposing hidden assumptions in everyday system attributions
2	Formal translation	Mapping intuitive boundaries onto the Markov Blanket formalism
3	Argumentative construction	Building and critiquing the life-mind continuity thesis
4	Conceptual analysis	Applying dynamical identity to classical puzzles
5	Position defense	Articulating and defending a philosophical position under objection